

Unit 2 Progress Check: MCQ

1. On the basis of electronegativity differences between atoms, which of the following scientific claims is the most accurate regarding the bonding in $\text{Mg}(\text{NO}_3)_2(s)$?

(A) There is polar covalent bonding between **Mg** atoms and **N** atoms.

(B) There is polar covalent bonding between **Mg** atoms and **O** atoms.

(C) There is ionic bonding between N^{5+} ions and O^{2-} ions.

(D) There is ionic bonding between Mg^{2+} ions and NO_3^- ions. ✓

2. Which of the following claims about a binary compound in which the bonding is ionic is most likely to be scientifically valid?

(A) Both elements in the compound are metals.

(B) The atomic masses of the elements in the compound are relatively small.

(C) There is equal sharing of electrons between the atoms of the elements in the compound.

(D) The electronegativity difference between the elements in the compound is relatively large. ✓

3. Which of the following claims about a binary compound composed of elements with the same electronegativity is most likely to be true?



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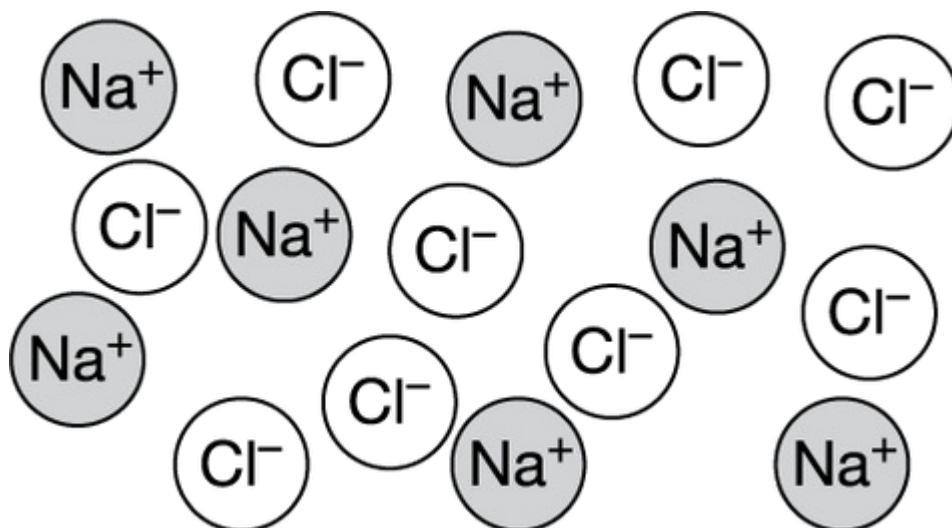
(A) The compound has properties similar to those of both elements.

(B) The bonding in the compound is nonpolar covalent. ✓

(C) The boiling point of the compound is above 1000°C .

(D) The compound contains strong ionic bonds.

4.



Molten (liquid) NaCl is represented by the particulate diagram shown above. Which of the following indicates whether $\text{NaCl}(l)$ conducts electricity and best explains why or why not?

(A) It conducts electricity because Na is a metal.

(B) It conducts electricity because ions are free to move. ✓

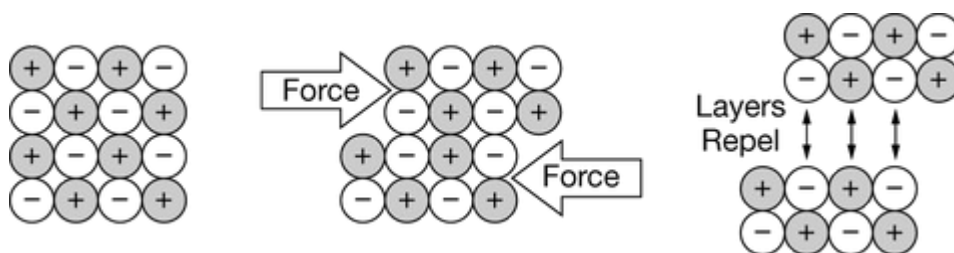
(C) It does not conduct electricity because Cl is a nonmetal.

(D) It does not conduct electricity because there are no free electrons.



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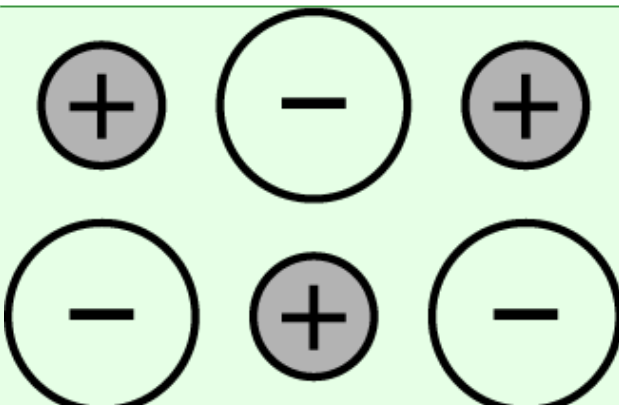
The particulate-level diagram shown above best helps to explain which of the following properties of ionic solids?

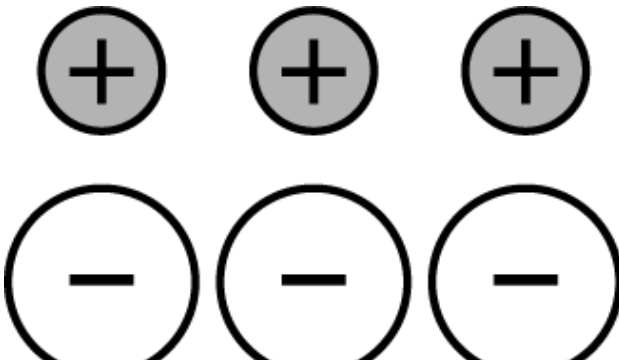
- (A) Density
- (B) Brittleness
- (C) Malleability
- (D) Conductivity

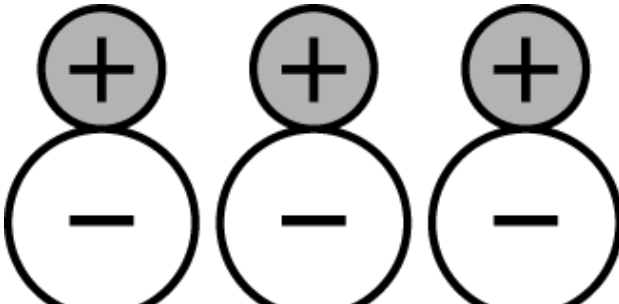
6. The particles in solid **KI**, a stable ionic compound, are arranged to maximize coulombic attractions while minimizing coulombic repulsions among the particles. Which of the following diagrams best represents the structure of solid **KI**?

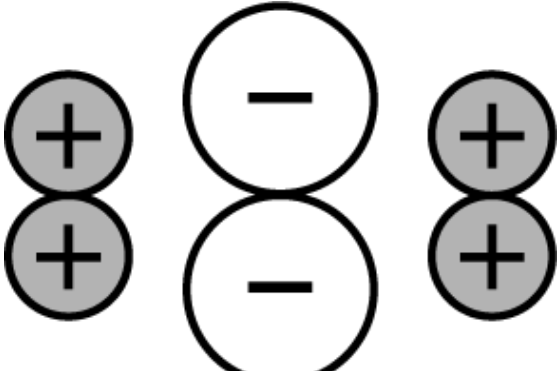


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(A)  

(B) 

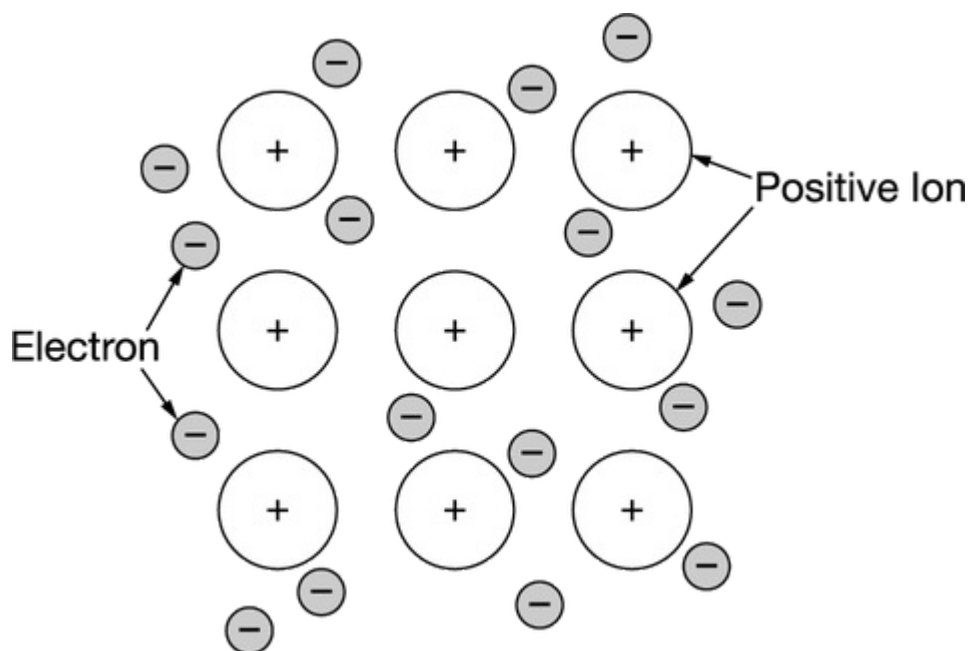
(C) 

(D) 



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7.



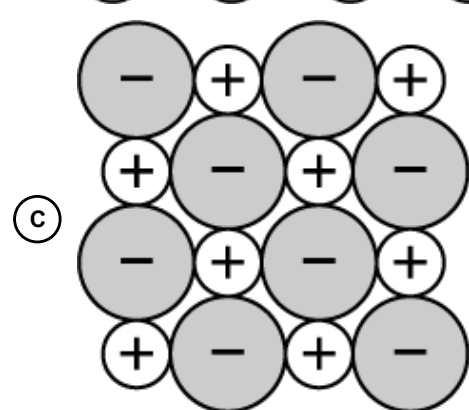
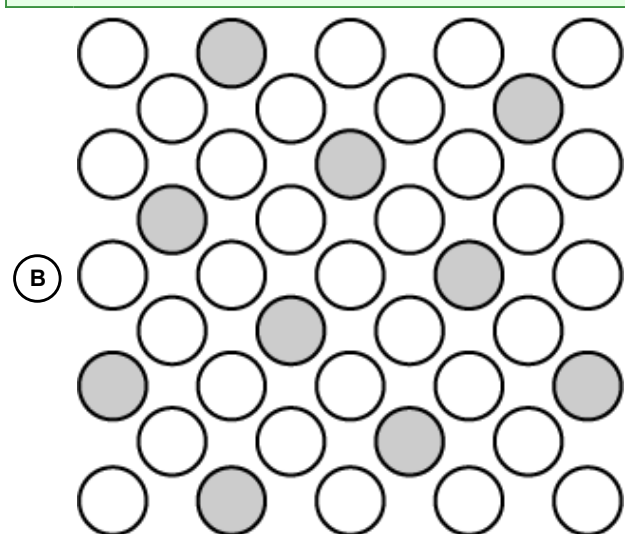
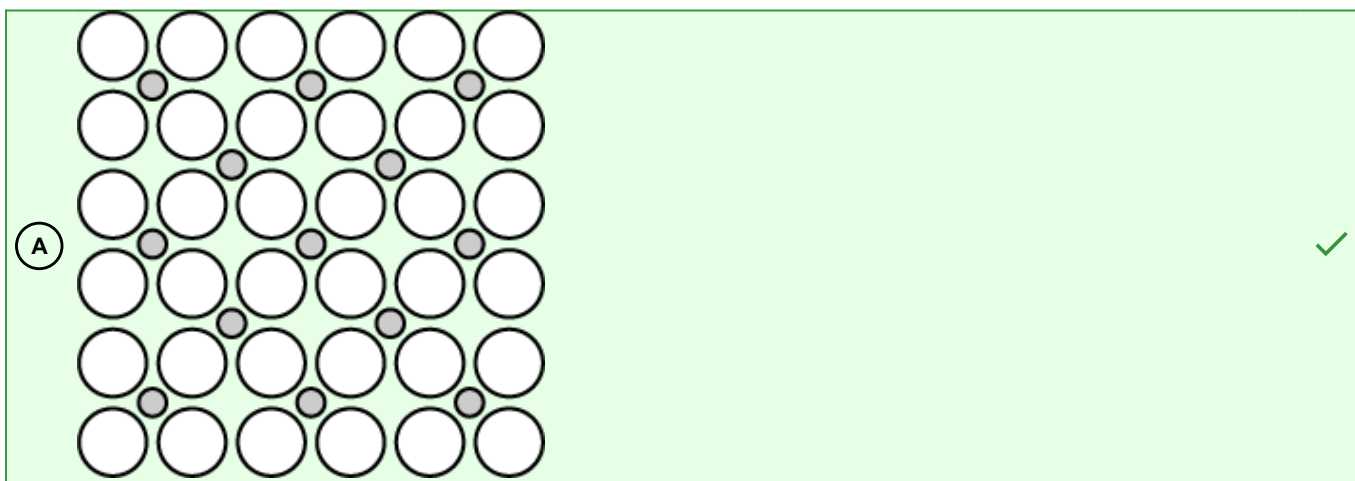
Which of the following indicates whether the solid substance represented by the particulate diagram shown above conducts electricity and explains why or why not?

- (A) It conducts electricity because it is made of positive and negative particles.
- (B) It conducts electricity because electrons are free to move through the substance. ✓
- (C) It does not conduct electricity because electrons are strongly attracted to specific positive particles.
- (D) It does not conduct electricity because the positive particles are not free to move through the substance.

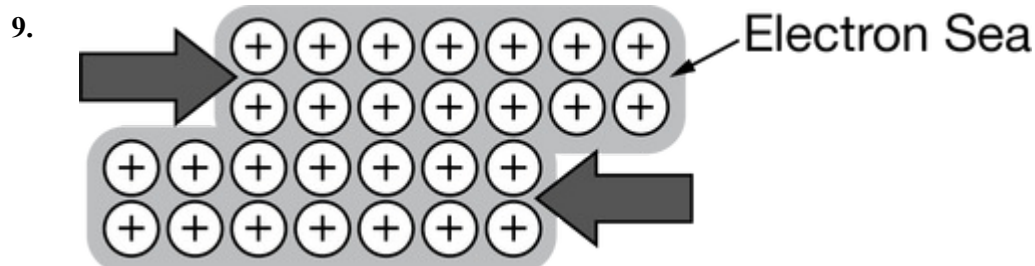
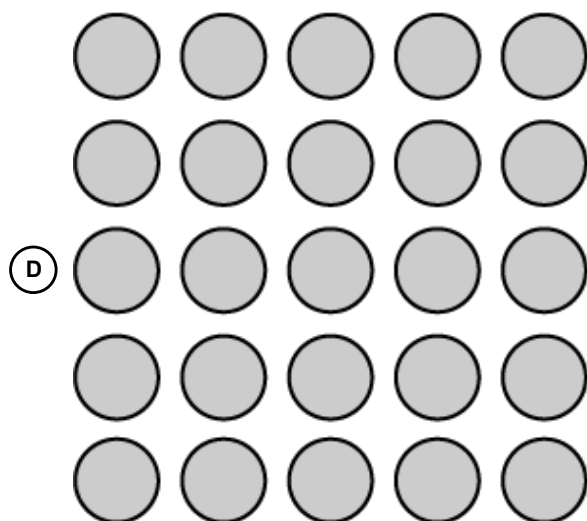
8. Which of the following particulate-level diagrams best represents an interstitial alloy?



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The diagram above best illustrates which of the following phenomena associated with solids that have metallic bonding?

- (A) Electrical conductivity, because it shows a lattice of positive ions immersed in a sea of electrons.
- (B) Malleability, because it shows how adjacent layers of positive ions can move relative to one another while remaining in full contact with the electron sea. ✓
- (C) Heat conductivity, because it shows how layers of atoms can slide past one another, creating friction between layers and causing the temperature of the solid to increase.
- (D) The ability to form substitutional alloys, because it shows how atoms of two different metals can combine in a one-to-one ratio.



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10.

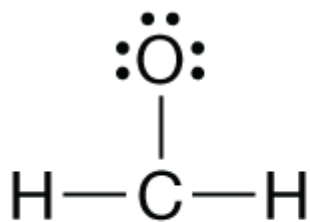


Diagram 1

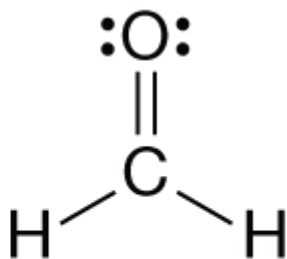


Diagram 2

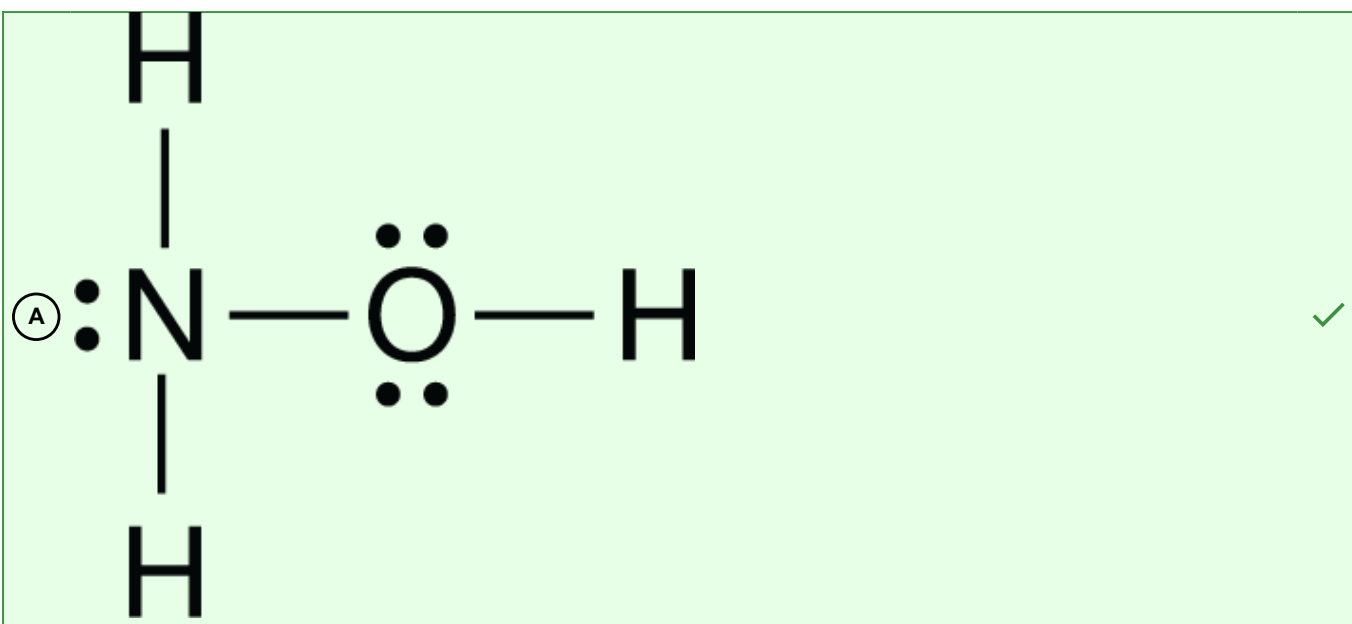
Which of the diagrams above best represents the CH_2O molecule, and why?

- (A) Diagram 1, because all bond angles are 180° .
- (B) Diagram 1, because all atoms have a formal charge of 0.
- (C) Diagram 2, because the molecule has a trigonal pyramidal shape.

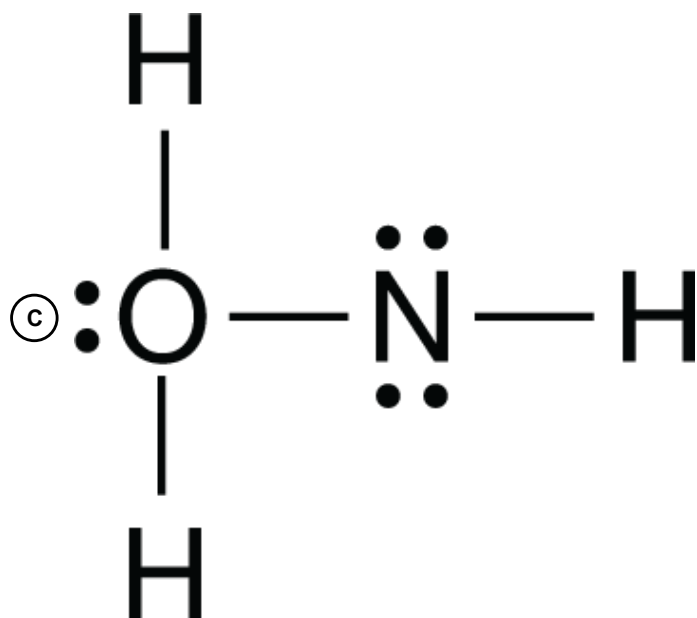
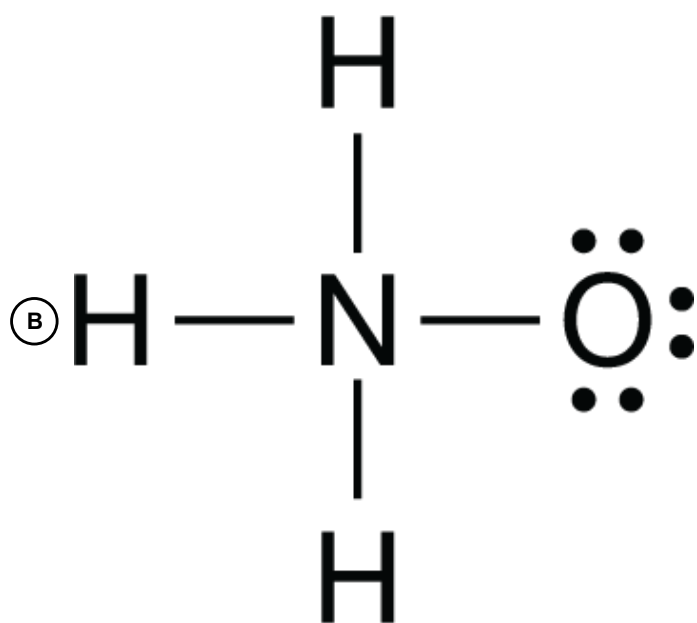
(D) Diagram 2, because all atoms have a formal charge of 0.



11. Based on formal charges, which of the following is the best Lewis electron-dot diagram for H_3NO ?



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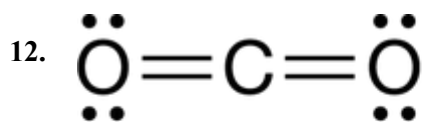
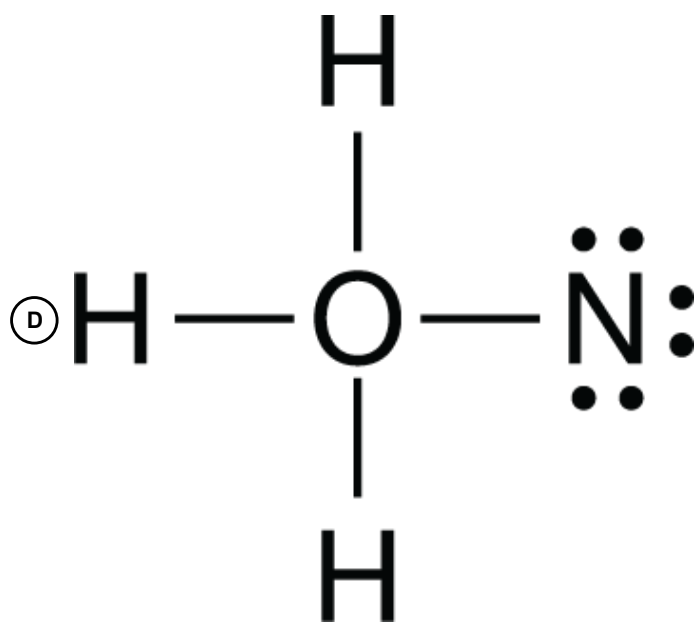


Diagram 1

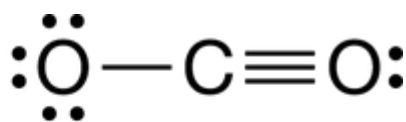


Diagram 2

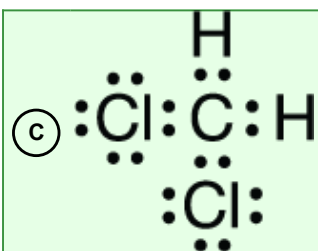
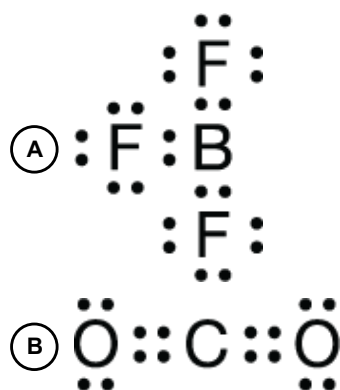
Which of the Lewis diagrams shown above is the more likely structure of CO_2 , and why?

- (A) Diagram 1, because all the atoms have a formal charge of 0. ✓
- (B) Diagram 1, because double bonds are stronger than triple bonds.
- (C) Diagram 2, because all the atoms have a formal charge of 0.
- (D) Diagram 2, because triple bonds are stronger than double bonds.

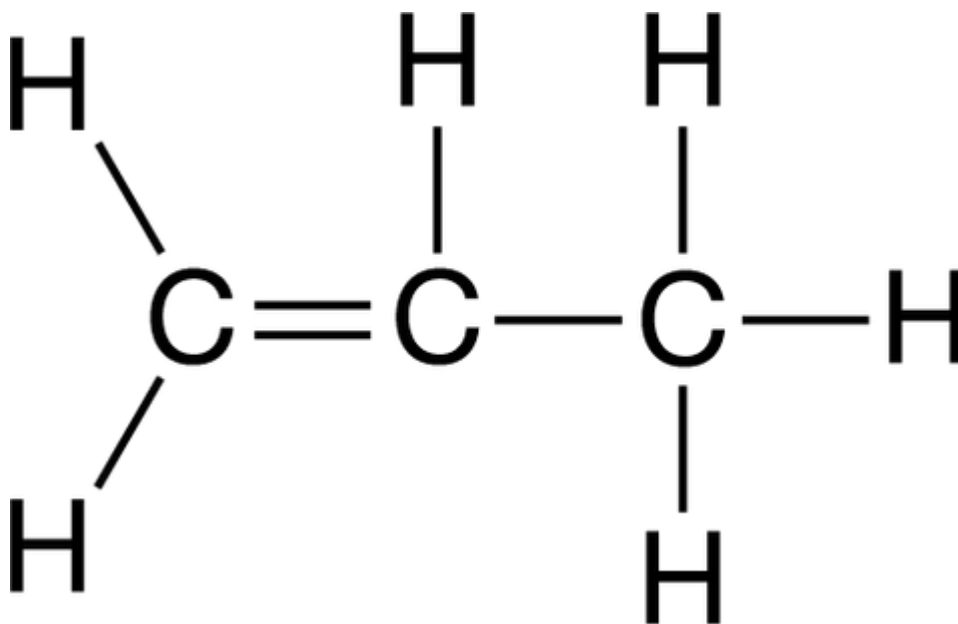
13. Which of the following Lewis diagrams represents a molecule that is polar?



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14.



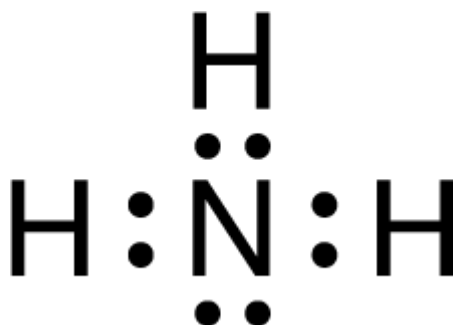
Which of the following correctly compares the strength of the two carbon-to-carbon bonds in the molecule represented in the Lewis diagram shown above?



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- ☒ A The carbon-to-carbon bond on the left is stronger because it is a double bond. ✓
- ☐ B The carbon-to-carbon bonds are the same strength because the C-C-C bond angle is 180° .
- ☐ C The carbon-to-carbon bonds are the same strength because they are both bonds between carbon atoms.
- ☐ D The carbon-to-carbon bond on the right is stronger because there are more hydrogen atoms attached to the rightmost carbon atom than to the leftmost carbon atom.

15.



Based on the Lewis diagram for NH_3 , shown above, the H-N-H bond angle is closest to which of the following?

- ☐ A 60°
- ☐ B 90°
- ☒ C 109.5° ✓
- ☐ D 120°